MIL-E-1/915B
10 March 1976
SUPERSEDING
MIL-E-1/915A
27 September 1968

MILITARY SPECIFICATION SHEET

ELECTRON TUBE, RECEIVING

TYPE 1616

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Rectifier, half-wave, high-vacuum

Outline	---	See figure 1
Base	---	A4-10
Cap	---	C1-5
Envelope	---	T16
Cathode	---	Coated filament

Base connections:

Pin No.	---	1	2	3	4	Cap
Element	---	f	nc	nc	f	a

ABSOLUTE-MAXIMUM RATINGS:

Parameter:	Ef	Epp	epx	ib	Io	i(surge)	Al
Unit:	Vac	Vac	kv	ma	mAdc	a	ft
Maximum:	2.75	---	6.0	800	130	2.5	10,000
Minimum:	2.25	---	---	---	---	---	---
Maximum:	2.75	---	7.0	350	15	0.3	10,000
Minimum:	2.25	---	---	---	---	---	---
TEST CONDITIONS:	2.50	2,250	---	800(min)	---	---	---

GENERAL:

Qualification - Required

Ⓑ denotes changes

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS		UNIT
							MIN	MAX	
	Quality conformance inspection, part 1								
1301	Filament current	-		0.65	II	If	4.6	5.4	A
1231	Emission	-	Eb = 150 Vdc	0.65	II	Is	275	---	mAdc
1353	Operation of rectifiers	1,2	t = 180	0.65	II	Io	260	---	mAdc
1201	Short and discontinuity detection	-		0.4	II	---	---	---	---
---	Visual and mechanical inspection criteria	-		---	---	---	---	---	---
---	Dimensions	-	See figure 1	1.0	I	---	---	---	---
	Quality conformance inspection, part 2								
1036	Bump	-	Angle = 20°	---	---	---	---	---	---
1101	Secureness of base, cap, or insert	-		---	---	---	---	---	---
1111	Base pin solder depth	-		---	---	---	---	---	---
(B)1211	Insulation of electrodes	4	Ef = 0; Eb = -500 Vdc (nominal)	4.0	S3	R	10	---	Meg
1105	Permanence of marking	-		---	---	---	---	---	---
	Quality conformance inspection, part 3								
---	Life test	-	Group C; epx = 5.5 kv; Rp/Io = 130 mAdc	---	---	---	---	---	---
---	Life-test end point: (500 hours):								
1231	Emission	-		---	---	Is	175	---	mAdc
(B)---	Dimensions	3	See figure 1	---	---	---	---	---	---

NOTES:

1. Use single-phase, full-wave circuit; apply anode voltage and filament voltage simultaneously to cold tube.
2. This test to be performed at the conclusion of the holding period.
(B) 3. *This test shall be performed during the initial production and once each succeeding 12-calendar months in which there is production. A regular double sampling plan shall be used, with the first sample of three tubes with an acceptance number of zero, and a second sample of three tubes with a combined acceptance number of one. In the event of failure, the test will be made as a part of quality conformance inspection, part 2, code level D, with an AQL of 6.5. The regular "12-calendar month" double sampling plan shall be reinstated after three consecutive samples have been accepted.*
(B) 4. Preheat with Ef = 2.5 Vac for 2 minutes (min). Remove filament power and read insulation resistance immediately.

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Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Army - MI, MU
Air Force - 17, 80
DSA - ES

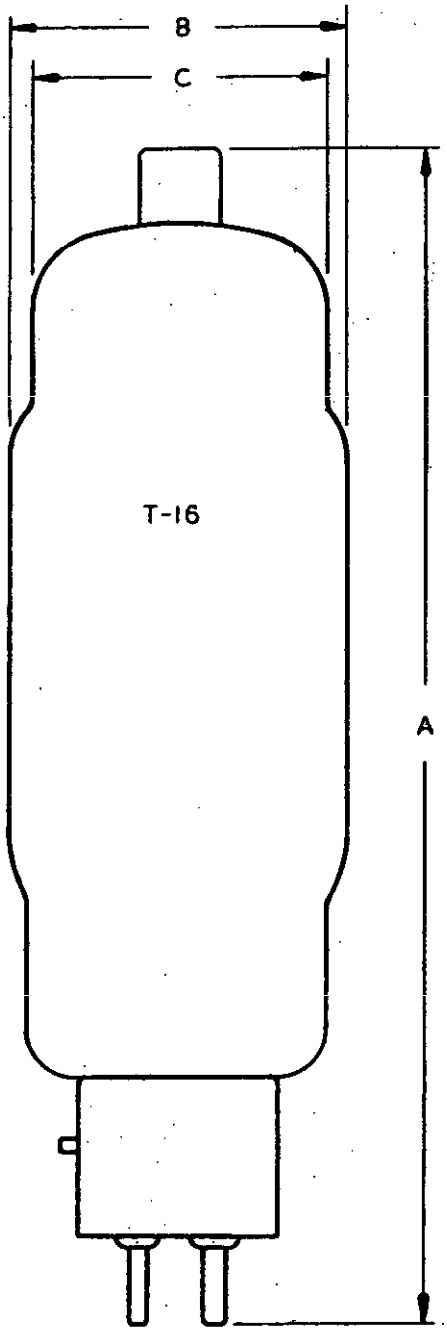
User activities:
Army - ME
Navy - AS, OS, MC, CG
Air Force - 11

Preparing activity:
Navy - EC

Agent:
DSA - ES

(Project 5960-3048)

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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Quality conformance inspection, part 1		
A	---	6.812 (173.02)
Quality conformance inspection, part 3 (see note 3)		
B	---	2.062 (52.37)
C	---	1.750 (44.45)

FIGURE 1. Outline drawing of electron tube type 1616.

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